Swinburne University of Technology
School of Science, Computing and Emerging Technologies

SWE30003

Software Architectures and Design

Assignment 2 – Object Design

Semester May, 2026
Worth 20 marks

Due:

<u>Electronic submission</u> (via Canvas)

<u>Expectation:</u>

Given the case study (same as in Assignment 1), you are expected to come up with an (initial/high-level) object-oriented design for the *case study system*. More specifically, you are expected to provide the following information:

<u>A list of classes</u> that define your solution as well as graphical representation of their inter-relationships (e.g., a basic UML class diagram), following the *Responsibility Driven Design* approach. *Do not include* any method names and/or signatures or attributes/fields as it would most likely be too premature to come up with this information for this initial design stage. A brief justification of your choice(s) must be given.
Note: if your design contains any record or struct-like abstractions, they must be defined as *data-holder classes* and included in your class diagram (or similar). But, note that they are not regarded as "normal classes" as they are more like "data structure definitions".

Similarly, considerations of user interfaces and database design should not normally form part of this initial/high-level OO design.

<u>For each of the identified classes, a CRC card</u> that gives a brief description of this class, all its responsibilities, and for each responsibility, all required collaborators. Please make sure that you list collaborators *on a per-responsibility basis*!

A particular requirement for individual group members is that each member of your assignment group must be mainly responsible for the description of at least two (2) classes / CRC-cards (while these CRC cards should be identified and reviewed by the whole group). The specific CRC cards concerned **must be identified in the contribution document** (see below). **A member failing to do so will incur deduction of that part's team mark AND achieve a 50% mark only for the entire assignment (at maximum)**.

An illustration/account of any *Design Patterns* or *Design Heuristics* used in your initial design, including a brief justification why they were used.

Note: As appropriate, you need to consider the use of any known Design Patterns and Design Heuristics, and such usage must be appropriately documented.

An illustration/account of the boot-strap (or initialisation) process of the *case study system*, i.e., which classes are responsible to create instances of which other class(es) and in which order.

An illustration/account of *four* typical, *non-trivial* use scenarios as verification of your design (e.g., how the system determines whether an input given by one of the user is valid and if so how it is applied to the specific business activities/process).

Wrap all the above up in a suitable document structure that contains authorship, a document overview, any assumptions you made, as well as any used external references. Please refer to the marking guide for further details. ***Please also attach your assignment 1 submission as an appendix.***

Please use the dedicated discussion forum on Canvas for any further clarifications for benefits to all, and ***no individual emails in this regard will be replied to***.

Note: The design you come up with for this assignment is to be used as the basis for a detailed design and implementation in Assignment 3. As such, it is important that *every member of your team fully understands this high-level solution design* that you come up with in this assignment!

Submission details:

Each assignment group is to submit their proposal (design) in **electronic form** through Canvas, along with the appropriately signed and completed "**Assignment and Project Cover Sheet**" declaration form, *which must be signed by all group members*. Each group is further required to submit a **contribution and collaboration document**, signed by *all* group members, which

1. lists the amount of time spent by each member on each significant part of the assignment,
2. describes briefly the specific contributions made by each group member, and
3. provides evidence showing that the assignment is done through *true* group collaboration, e.g., discussions and mutual reviews of **all** major parts of the assignment, and simple group meeting minutes (time, place, attendees, issues discussed, decisions made, etc).

Note: For the assignments in this Unit of Study, students are to work in groups of three or four, i.e., the same groups as in Assignment 1. Permission by the Unit of Study convener is required to *change groups*, **well before** the submission deadline, with a "good" reason. Extensions to the submission deadline can only be granted for genuine reasons and the Unit of Study convener must be contacted *at least 48 hours* prior to the submission deadline.

The **electronic submission** is through Canvas by the deadline as published in the front of this assignment specification.

The Unit of Study convener reserves the right to call in any assignment teams to further explain their submission if there are doubts about the authorship of the presented solution.

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ASSIGNMENT AND PROJECT COVER SHEET

Subject Code: SWE30003 Unit Title: Software Architectures and Design

Assignment number and title: 2, Object Design Due date: 11:59pm,12th Oct 2025

Tutorial Day and time: ____________________ Project Group: _______

Tutor: ________________

To be completed as this is a group assignment
We declare that this is a group assignment and that no part of this submission has been copied from any other student's work or from any other source except where due acknowledgment is made explicitly in the text, nor has any part been written for us by another person.

ID Number	Name	Signature

Marker's comments:

Total Mark:________________

Extension certification:

This assignment has been given an extension and is now due on ________________________

Signature of Convener:____________________